

1. Quiz (15 Questions – Mix of MCQ, True/False, Short Answer)

Concept Check – Java Object & Class

Part A: Multiple Choice (5 questions)

1. What does the **new** keyword do in Java?
 - A. Declares a new class
 - B. Creates a new method
 - C. Creates a new object from a class
 - D. Deletes an object
2. Which of the following is a valid Java constructor for a class named **Dog**?
 - A. `public void Dog()`
 - B. `public Dog()`
 - C. `void Dog()`
 - D. `constructor Dog()`
3. What is the default return type of a constructor?
 - A. `int`
 - B. `void`
 - C. `String`
 - D. No return type
4. What is the output of the below code ?

```
class Constructor
{
    static int x = 1;
    public Constructor()
    {
        x = 5;
    }

    public void showX()
    {
```

```
        System.out.println("x : " + x);
    }

    public static void main(String[] args)
    {
        Constructor obj = new Constructor();
        System.out.println(x);
        obj.showX();
    }
}
```

A. 5

x : 5

B. x : 5

x : 5

C. **Compilation Error**

D. **Run time error**

5. **Which of these is NOT a benefit of OOP?**

- A. Code reuse
- B. Scalability
- C. Better security
- D. Slower performance

Part B: True/False (5 questions)

- 6. A Java class can have multiple constructors.
- 7. Objects in Java are created from classes.
- 8. You must always define a `main()` method inside a class to compile it.
- 9. Instance variables are declared inside methods.

10. The `this` keyword refers to the current object.

Part C: Short Answer (5 questions)

11. Define what an object is in Java.

12. What is the difference between a class and an object?

13. Why do we use constructors in a class?

14. Write the code to create an object of `Car` class.

15. Name two things that a class can contain.



2. Code Examples



Example: `Student` Class

Unset

```
public class Student {
    String name;
    int age;

    public Student(String studentName, int studentAge) {
        name = studentName;
        age = studentAge;
    }

    public void printDetails() {
        System.out.println("Name: " + name + ", Age: " + age);
    }

    public static void main(String[] args) {
```

```
        Student s1 = new Student("Alya", 20);
        s1.printDetails();

        Student s2 = new Student("Zafran", 22);
        s2.printDetails();
    }
}
```

Challenge Task: **Book Class** (Student Practice-Fix the errors)

Unset

```
public class Book {
    String title;
    String author;
    int yearPublished;

    public Book(int y, String a, String t) {
        title = t;
        author = a;
        yearPublished = y;
    }

    public void displayBookInfo() {
        System.out.println("Title: " + title + ", Author: " +
author + ", Year: " + yearPublished);
    }

    public void isClassic() {
        return yearPublished < 2000;
    }

    public static void Main(String[] args) {
        Book 1b = new Book("To Kill a Mockingbird", "Harper Lee", 1960);
        displayBookInfo();
    }
}
```

```
        System.out.println("Classic? " + isClassic);
    }
}
```

Check and Fix the errors in Book.java . The output should be as below :

Output:

```
Title: To Kill a Mockingbird, Author: Harper Lee, Year: 1960
Classic? true
```

3. Student Worksheet

Title: Object Class in Java - Practice Worksheet

Section A – Fill in the Blanks

1. A _____ is a blueprint for creating objects.
2. An _____ is an instance of a class.
3. The _____ keyword is used to create objects.
4. A _____ has the same name as the class and no return type.
5. Use the `this` keyword to refer to the _____ object.

Section B – Write Code

Create a `Movie` class with:

- Fields: `title`, `genre`, `rating` (double)
- Constructor
- Method: `printMovieInfo()`

Then create 2 `Movie` objects in `main()` and call the method.

Section C – Debug the Code

Find and fix the errors:

Unset

```
public class Car {
    String model;
    int year;

    public void Car(String m, int y) {
        model = m;
        year = y;
    }

    public void printInfo() {
        System.out.println("Model: " + model + ", Year: " +
year);
    }

    public static void main(String[] args) {
        Car myCar = new Car("Honda", 2022);
        myCar.printinfo();
    }
}
```

Section D – Reflection (Short Answer)

1. What was easy about today's coding tasks?
 2. What was hard or confusing?
 3. Rate your confidence (1–10) with creating Java objects.
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